

Caleb Elder

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Education

University of Georgia

December 2026

3.98 GPA | B.S. Computer Science, Minor in Mathematics

Achievements: Hollingsworth Award in Linear Algebra, Presidential Scholar, Dean's List, Zell Miller Scholarship

Relevant Coursework: Algorithms, Data Structures, Graph Theory, Applied Linear Algebra, Discrete Mathematics, Systems Programming, Computer Architecture, Computer Networks, Theory of Computing, Mathematical Analysis, Software Engineering, Software Development

Technical Skills

Programming Languages (Proficient): Java, Python, C

Programming Languages (Familiar): TypeScript, JavaScript, SQL, C++, HTML/CSS, LaTeX

Cloud & Developer Tools: Azure (App Service, Functions, ACI), Git, GitHub Actions (CI/CD), Docker, Linux, Bash

Frameworks & Libraries: React, Node.js, Express, Supabase (Postgres RLS), Playwright, NumPy, Pandas, PyTorch

Data & Protocols: REST APIs, MySQL, PostgreSQL, JSON, Gson, MSAL/Azure AD Auth

Human Languages: Proficient: Spanish | Intermediate: French, Chinese | Conversational: German, Russian, Korean

Experience

Fetch Freight

May 2026 – Present

Software Engineering Intern

Birmingham, AL

- Shipped 12 production web apps and serverless services (TypeScript, React, Node.js, Azure), including an AI document engine (Claude Vision) processing 300+ daily invoices and draining a 10-day backlog
- Engineered event-driven Azure Functions and Playwright/Docker browser workers for portal syncs and real-time SMS alerts, utilizing atomic claim-before-send ledgers to eliminate duplicate sends
- Strengthened platform security by designing signed single-use capability tokens for unauthenticated workflows, enforcing Postgres Row-Level Security, and resolving a vendor link flaw exposing internal data
- Reverse-engineered legacy SQL reports into modern web applications with custom parity-check harnesses to reduce TMS database load, while establishing heartbeat alerting to eliminate silent failures

AI4STEM (VIPR Program)

August 2025 – December 2025

Undergraduate Research

Athens, GA

- Researched LLM automated assessment models, sourcing benchmark datasets and designing a multimodal rubric integrating text and image data
- Developed Python data pipelines to process spreadsheet metrics into structured JSON, hitting the OpenAI API to generate automated, reasoning-based grading scores
- Evaluated LLM system performance using Cohen's Kappa, achieving 0.66-0.69 (substantial agreement) across 3 trials correlating model outputs against human teacher scores; presented findings at the CURO Symposium

Projects

AI Language Learning Platform | *Next.js, TypeScript, Supabase, Vercel AI SDK*

Summer 2026

- Built a full-stack Spanish/French learning platform (CEFR A1-C2) with AI stories, streaming tutoring, dictionary/conjugations, spaced-repetition vocab, and multiplayer games
- Implemented a streaming AI tutor (GPT-4o-mini) with abortable client streaming, session persistence, and background personalization that re-analyzes learner traits and pre-queues conversation starters
- Designed Supabase Postgres schema with row-level security and Realtime chat/pictionary rooms; added `after()` generation queues so stories and chat starters pre-generate without blocking responses
- Shipped structured AI pipelines with Zod schemas across story generation, grammar drills, pronunciation scoring (ElevenLabs + Web Speech), and an offline crossword seed generator with grid validation

Cinema E-Booking System | *React, TypeScript, Flask, MySQL, Docker*

Spring 2026

- Collaborated on a full-stack cinema booking platform with movie search, interactive seat selection, checkout, payments, promotions, admin portals, and email confirmations
- Built a typed React/TypeScript frontend against a Flask REST API and MySQL schema, containerized with Docker Compose for consistent team development
- Applied design patterns (Facade, Builder, Adapter, Proxy) to keep booking, payment, and admin flows modular and maintainable across a multi-developer codebase